

# Pharmaceutical manufacturing in the national economy

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A 42-country analysis of how large the pharmaceutical industry is inside each economy, how concentrated that exposure has become, and what each country produces.

Countries

**42**

Period

**1990 - 2025**

Sources

**Eurostat · OECD · trade statistics**

# Three findings

01

## **Scale is remarkably uniform.**

In every large economy, pharmaceutical manufacturing accounts for approximately 1% of output — and has done so for three decades.

02

## **Concentration is the real variable.**

Shares of 7–17% occur only in small economies where a single industry has come to dominate: Ireland, Denmark and Switzerland.

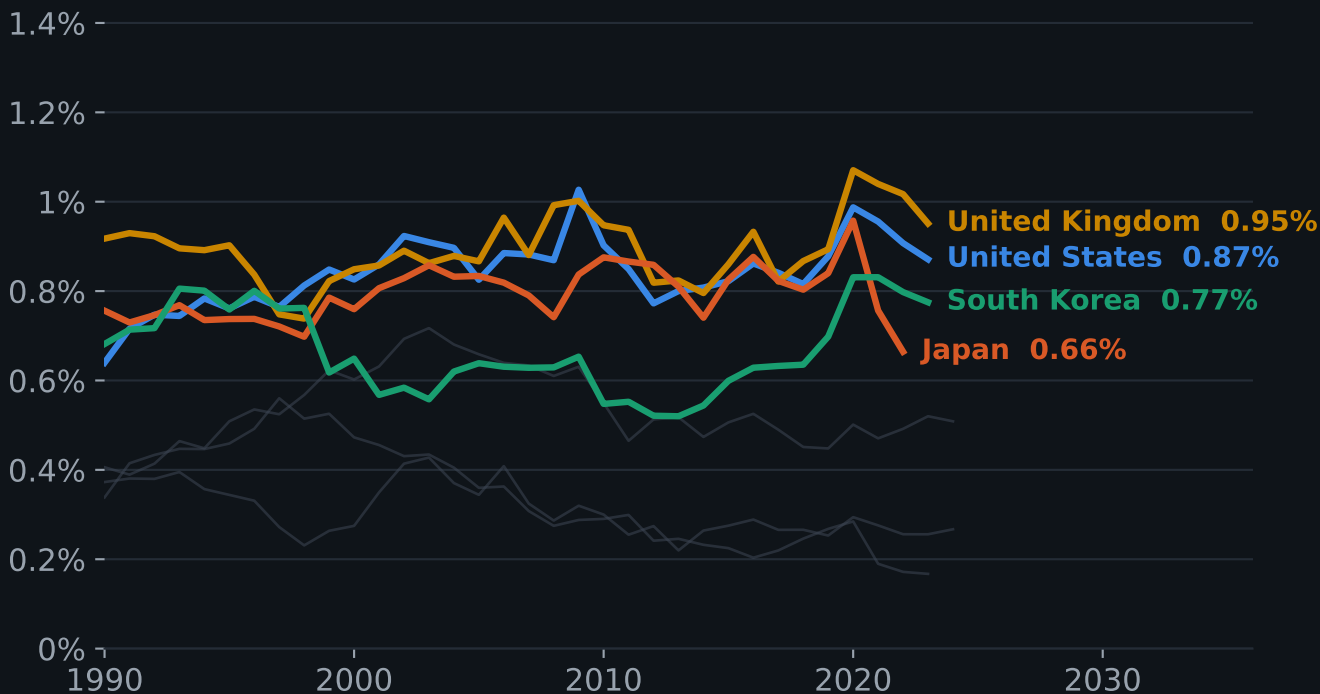
03

## **Composition differs more than scale.**

Trade data separates biologics producers from finished-dose manufacturers — a distinction invisible in the headline economic figures.

# Large economies hold pharmaceuticals near 1%

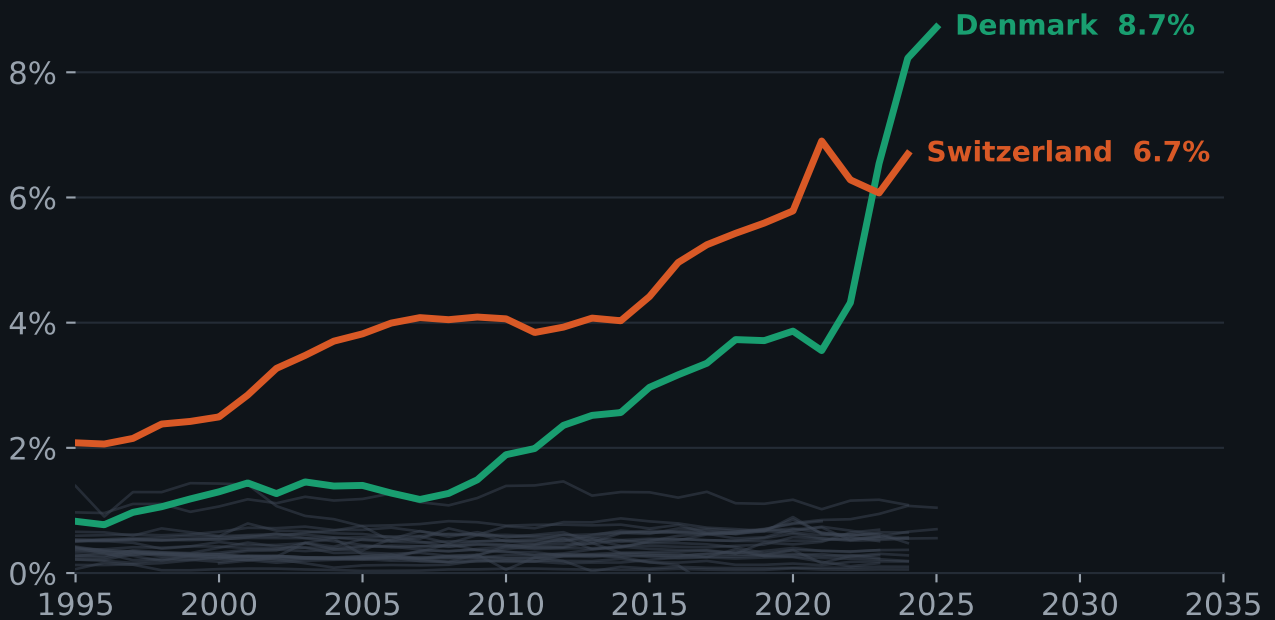
Share of total value added, 1990 to most recent year.



The band has held through the genomics era, the shift to biologics and the pandemic. Grey lines: Canada, Mexico, Australia, France, Italy, Spain, Germany.

# The exceptions are all small economies

Pharmaceutical share of GDP, 1995–2025.



In 2023 Germany and Denmark each produced roughly €25bn of pharmaceutical value added. That is 0.6% of the German economy and 6.5% of the Danish one. Concentration, not sector size, determines macroeconomic exposure.

# Two routes to concentration

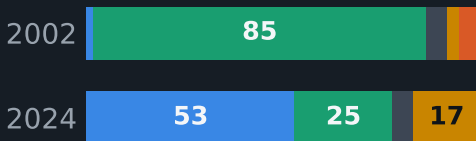
Ireland repositioned its output. Denmark deepened a single franchise. Both reached scale; the composition differs.

## Ireland

**6.3% → 16.7%**

of value added, 1995–2023

Exports €15bn → €99bn



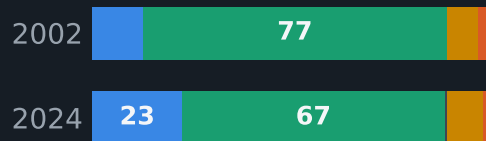
Export composition, %

## Denmark

**0.8% → 8.7%**

of GDP, 1995–2025

Exports €4bn → €22bn



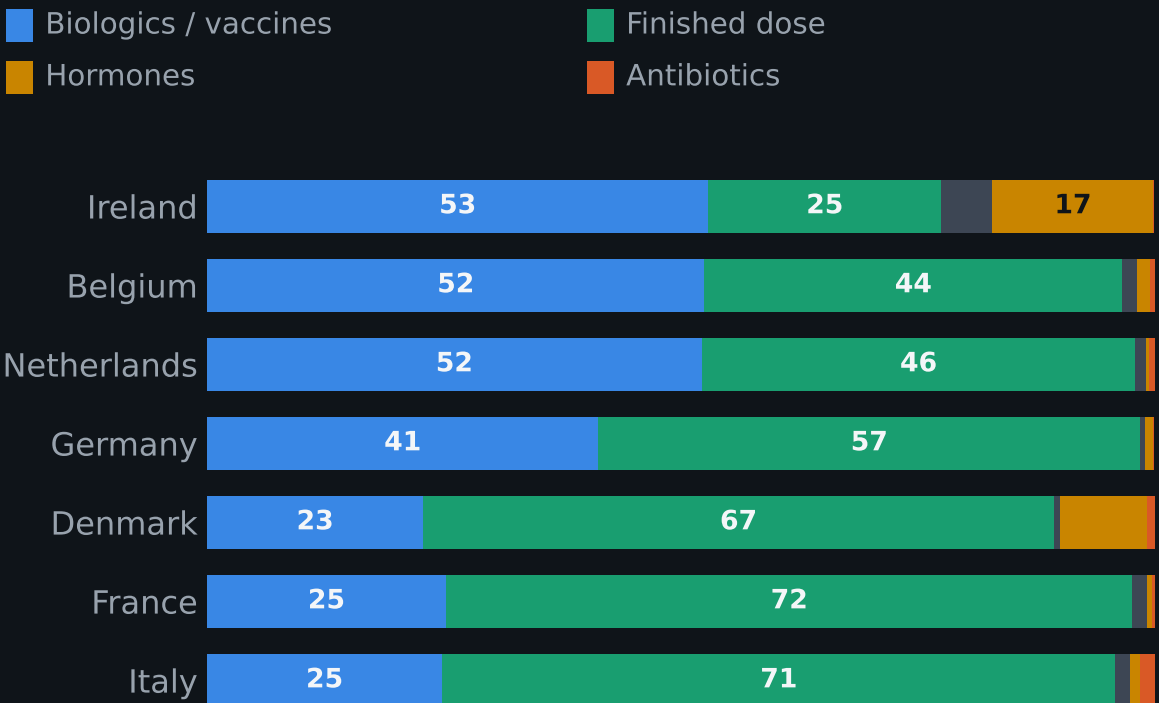
Export composition, %

■ Biologics / vaccines   ■ Finished dose   ■ Hormones

Ireland moved from 85% finished dose in 2002 to 53% biologics and 17% hormones in 2024 — a change of product, not only of volume. Denmark remained dose-led throughout, scaling a diabetes and obesity franchise rather than repositioning.

# Two industries share one label

Share of pharmaceutical exports by product class, 2024.



The same distinction holds outside Europe: 97% of South Korea's pharmaceutical exports to the EU are biologics, against 20% in 2013 — a national capability built within a decade.

# What the analysis supports

## Economic analysis

Danish, Irish and Swiss aggregates warrant an ex-pharmaceutical view before macro conclusions are drawn.

## Industrial strategy

Composition can be changed within a decade — Ireland and South Korea repositioned output rather than expanding it.

## Türkiye

0.35% of GDP, broadly unchanged since 2003 against a 0.5% median. The constraint is positioning, not data.

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Method · NACE/ISIC C21 value added, Eurostat and OECD STAN, in national currency. Composition from HS chapter-30 trade, a proxy for output. Full sources and caveats online.

[namikakmandev.github.io/pharma-gdp-share.html](https://namikakmandev.github.io/pharma-gdp-share.html)